

Labanya Saha

Computer Science Graduate

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RESEARCH INTERESTS

Artificial Intelligence – Machine Learning – Deep Learning – Computer Vision – Natural Language Processing – Multimodal Learning – Large Language Models (LLMs)

EDUCATION

- B.Sc. in Computer Science and Engineering 2022–2025 University of Asia Pacific Dhaka, Bangladesh
CGPA: **3.46/4.00**

Thesis Title: *Behind the Laughter: Uncovering Gender Bias in Code-Mixed Bangla Memes*

Thesis Supervisor: *Noor Mairukh Khan Arnob*

PUBLICATIONS

- **Saha, L., et al.** (2026). *Behind the Laughter: Uncovering Gender Bias in Code-Mixed Bangla Memes*. ACL Workshop (LT-EDI 2026).

[ACL Anthology](#).

Developed multimodal gender bias detection models using BanglishBERT and ConvNeXt with PyTorch, working with a multimodal Bangla meme dataset containing more than 7,000 image-text samples.

EXPERIENCE

- Intern July 2025 – Oct 2025 HerWILL Remote
 - Collected and organized more than 1,500 misogynistic social media comments for NLP-based analysis.
 - Authored four AI research articles covering healthcare AI, LLMs, and responsible AI.
 - Co-authored an AI handbook for school students introducing AI concepts and career guidance.
- Support System Executive July 2025 – Sep 2025 eChithi Hybrid
 - Managed and validated operational datasets for system accuracy and reporting.
 - Worked with cloud-connected IoT devices and real-time sensor data processing.
 - Collaborated with cross-functional teams during technical workshops and deployment activities.

RESEARCH PROJECTS

- **Audio Deepfake Detection: A Systematic Review** *Ongoing*
 - Conducting a systematic review of research on audio deepfake and synthetic speech detection.
 - Reviewing existing deep learning, speech representation, and large language model-based approaches for audio deepfake detection.
 - Analyzing datasets, evaluation methods, and research trends to identify challenges and potential directions for future research.
- **LLM-Assisted Medical Research** *Ongoing*
 - Exploring large language models and agent-based approaches for medical research using biomedical literature and structured datasets.
 - Analyzing and integrating information from research papers and domain-specific datasets to support systematic investigation.

- Investigating LLM-based approaches for information extraction, evidence organization, and research-oriented reasoning.
- **Computer Vision for Automated Mulberry Leaf Disease Classification** *Ongoing*
 - Investigating deep learning approaches for mulberry leaf disease classification using image data.
 - Studying model performance under variations in image quality and visual conditions.
 - Exploring controlled image perturbations including illumination changes, blur, noise, and resolution degradation.
 - Comparing baseline and robustness-oriented training approaches using transfer learning.
 - Implementing experiments using Python and PyTorch.

PROJECTS

- **Medicine-CHAI** – Django, PostgreSQL
Full-stack medical inventory management system with authentication, CRUD operations, and an administrative dashboard.
[GitHub Repository](#)
- **Helping & Connect People** – Django
Community volunteering platform with authentication, role-based access, and administrative moderation.
[GitHub Repository](#)

TECHNICAL SKILLS

Programming	Python, C++, C#, Java, JavaScript, SQL
Research Areas	Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Multimodal Learning
Machine Learning	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, BanglishBERT
Tools	Git, GitHub, LaTeX, Overleaf, Linux, Docker, VS Code, Jupyter
Web Technologies	Django, Flask, Node.js, React, REST APIs
Databases	PostgreSQL, MySQL
Languages	Bangla (Native), English

ACHIEVEMENTS

- Solved **1069+ programming problems** on CodeChef, Codeforces, and other online judges.
- **CodeChef: 3-Star**, highest rating **1624**, with participation in 80+ rated contests.
[CodeChef Profile](#)

CERTIFICATIONS

- **Django Application Development with SQL and Databases** – Coursera
- **Introduction to Web Development** – Coursera
- **Introduction to Machine Learning** – Research & Publication Unit, CSE-UAP

REFERENCES

Noor Mairukh Khan Arnob

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